

Business Report of Automobile Purchase

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1. Introduction

- The automobile industry is one of the most competitive sectors in the global market, where understanding customer behavior plays a crucial role in improving sales performance and marketing strategies. Companies rely on data-driven insights to understand the factors that influence customers' automobile purchasing decisions.
- This business report analyzes automobile purchase data to identify patterns and relationships between customer demographics, financial characteristics, and vehicle purchasing behavior. The dataset contains information about customers such as **age, gender, profession, income level, partner employment status, loan status, and automobile price**. By analyzing these variables, we aim to understand how different factors influence the choice of vehicle and the amount spent on purchasing automobiles.

2. Data Overview

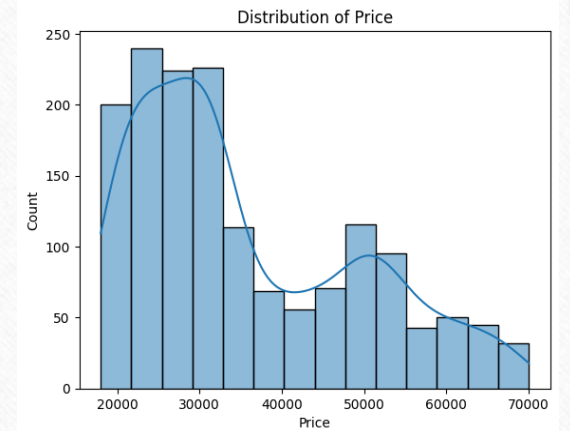
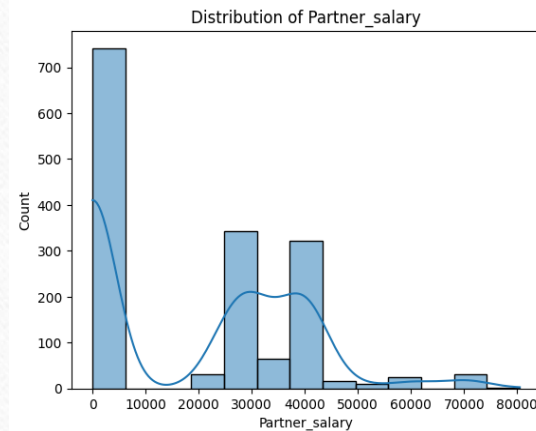
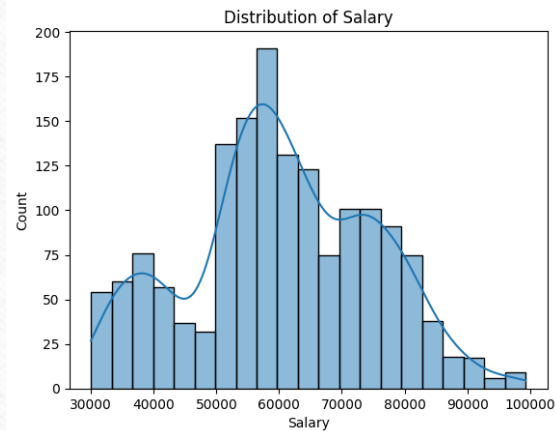
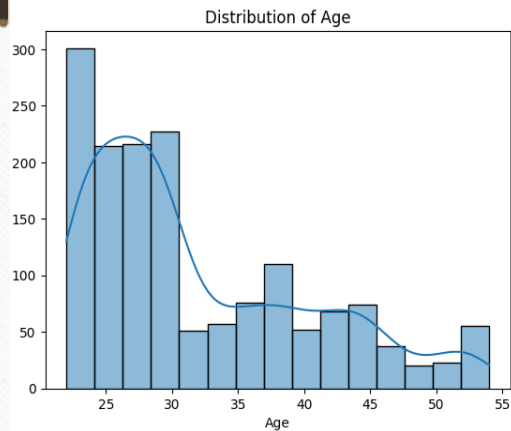
- This dataset contains total of 14 variables and 1581 customers/records.
- Dataset Description:

1. Gender : Male, Female	8. Total Salary
2. Age : 20 to 40 (1277), 40 to 60 (304)	9. House Loan
3. Education	10. Personal Loan
4.. Profession	11. Partner Working
5. Marital Status : Single (138), Married (1443)	12. Partner Salary
6. No of Dependents	13. Make (Car type)
7. Salary : \$0 to \$50,000 (323), \$50,000 to \$1,00,000(1258)	14. Price

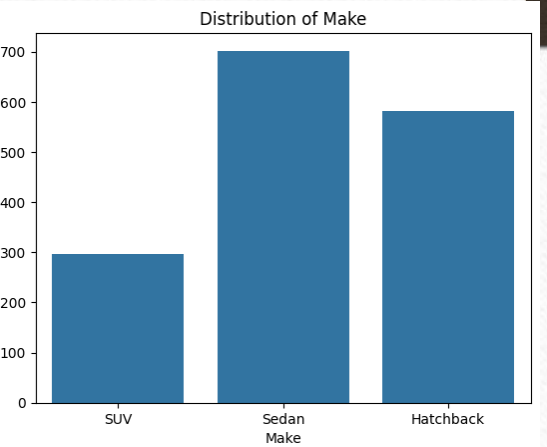
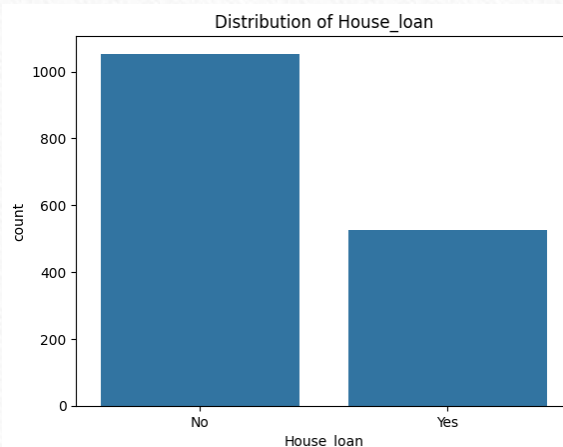
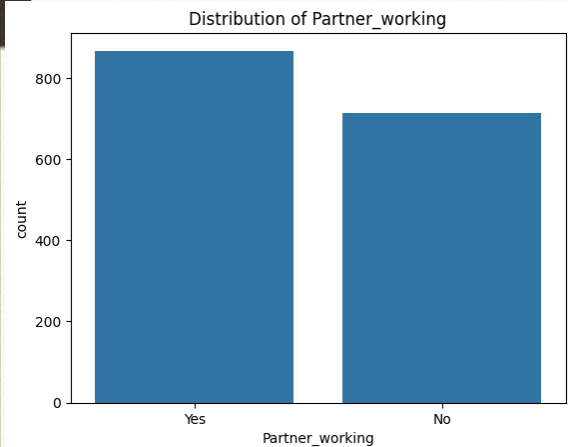
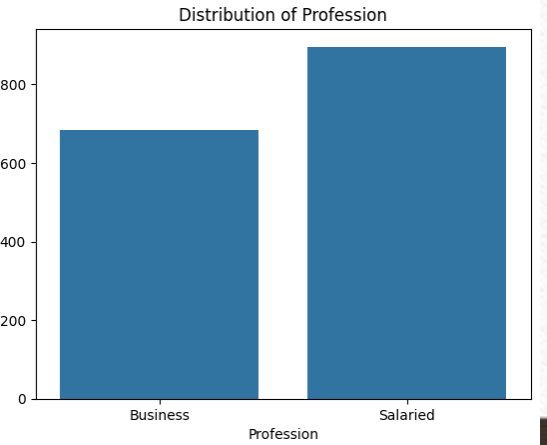
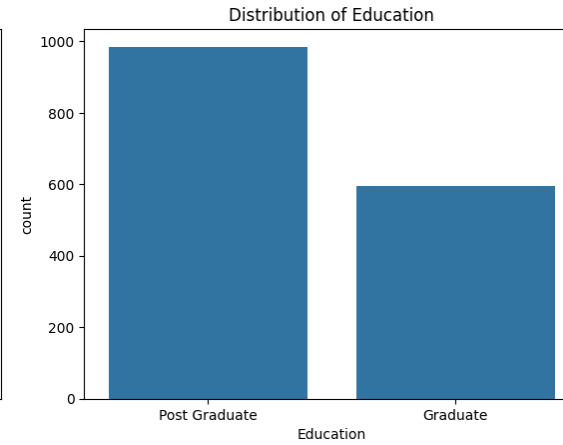
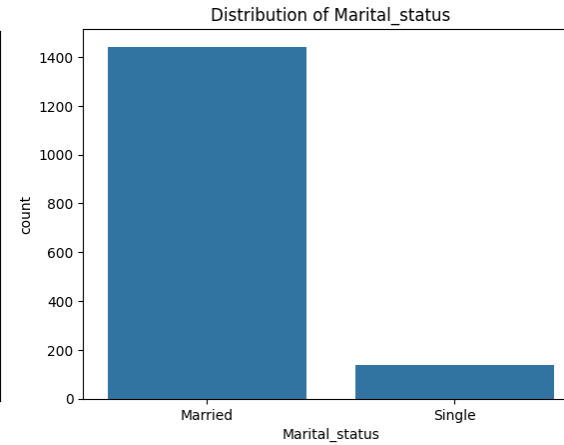
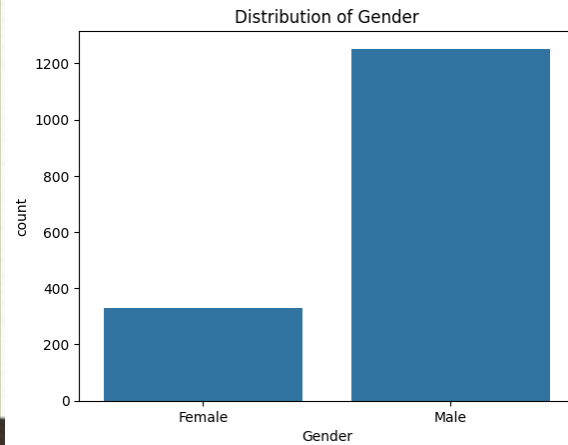
3. Univariate Analysis

- Univariate analysis examines the distribution of individual variables to understand customer demographics and vehicle purchase characteristics.

1. Numerical Variables:



2. Categorical Variables:

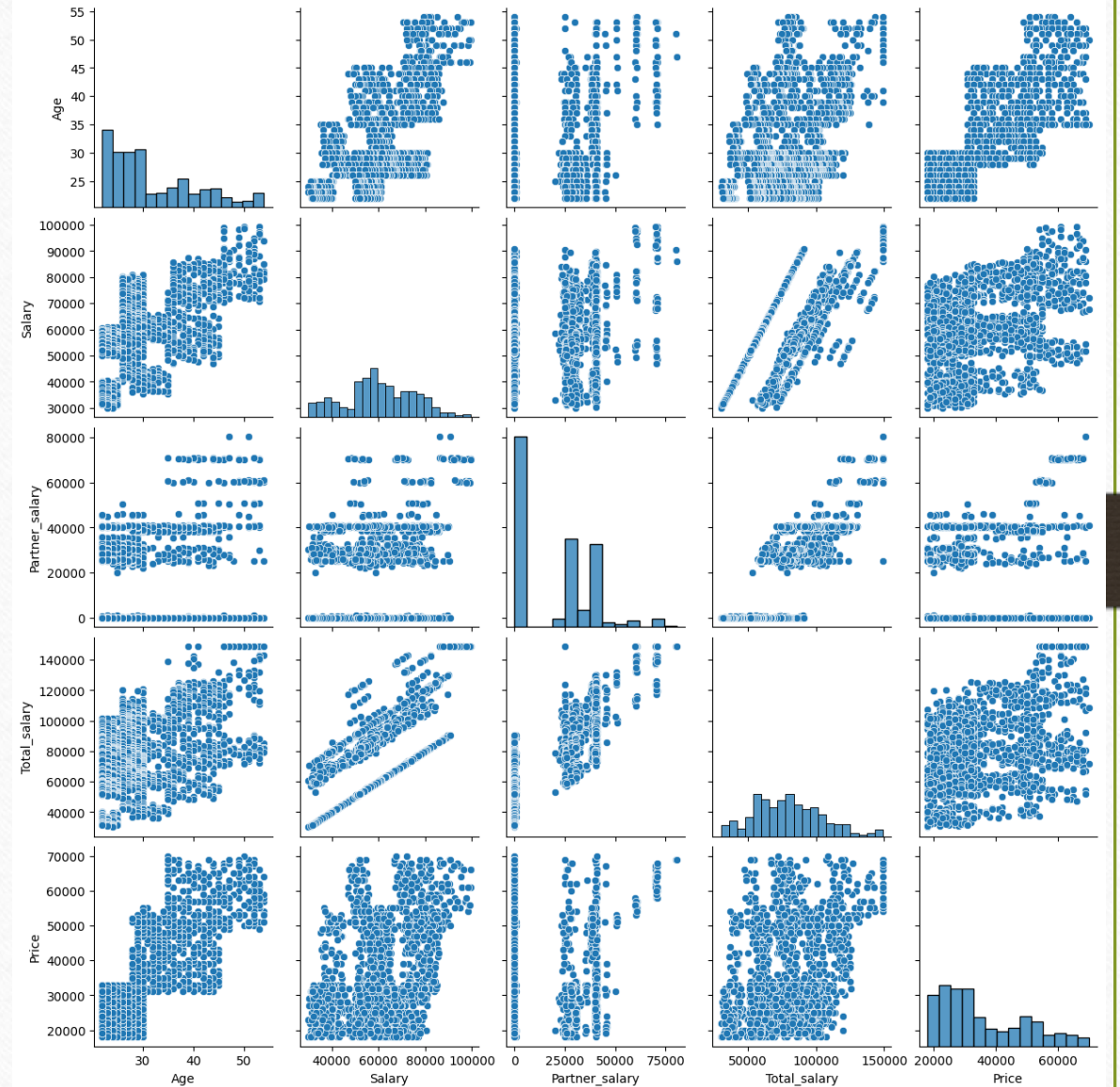


4. Bivariate Analysis

- Bivariate analysis explores relationships between variables such as gender and automobile price or partner employment and purchasing behavior.

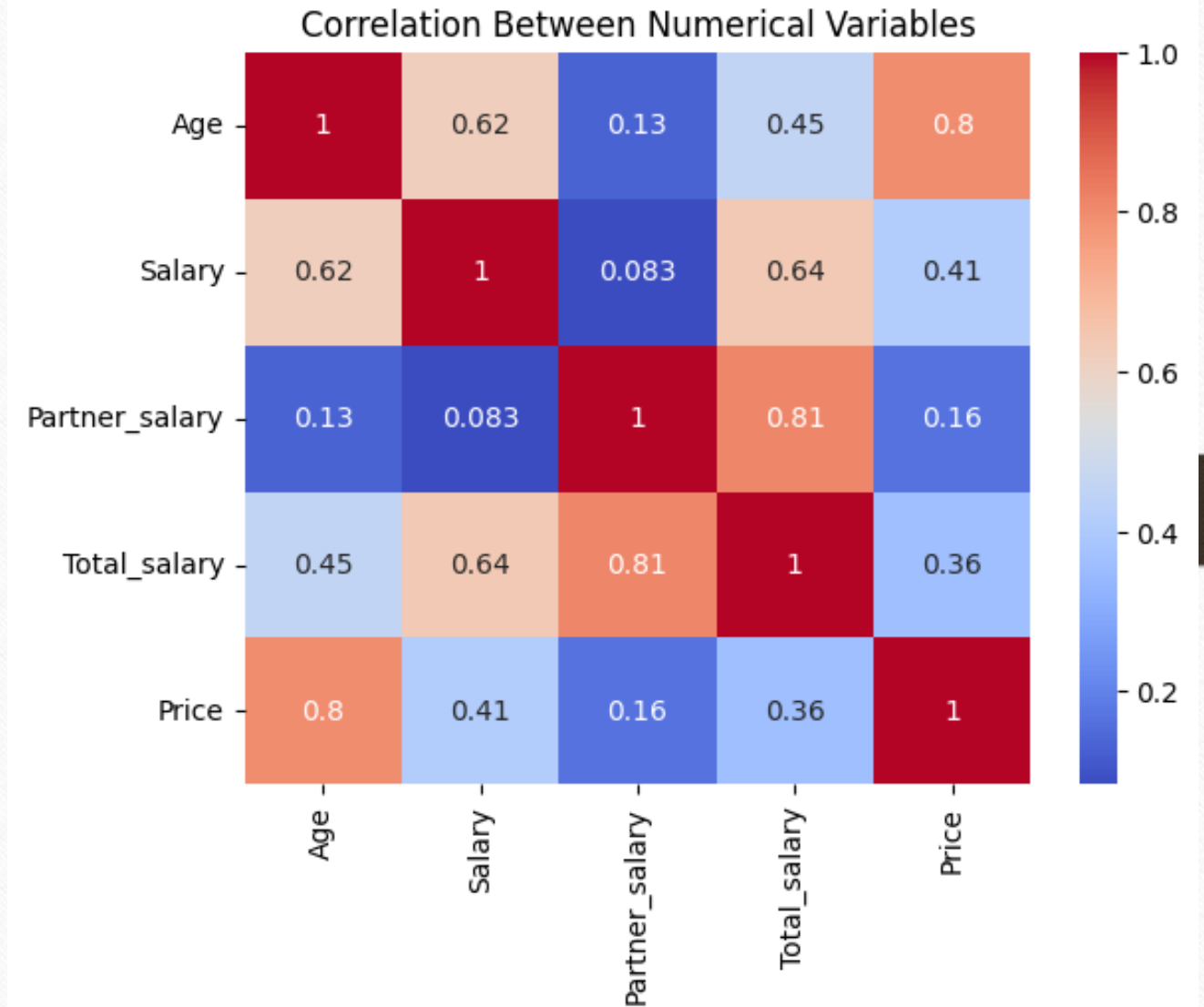
1. Relationship Between Numerical Variables:

- Total salary increases when Salary and Partner salary increase.
- Price tends to increase with Total salary, indicating wealthier households buy more expensive cars.
- Age shows a mild relationship with salary, as income generally rises with experience

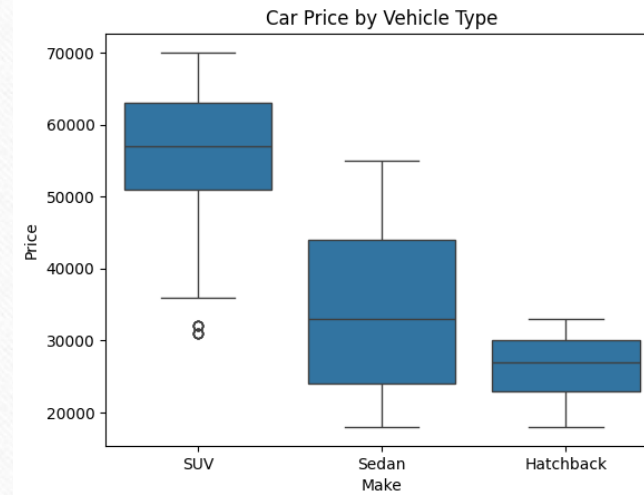
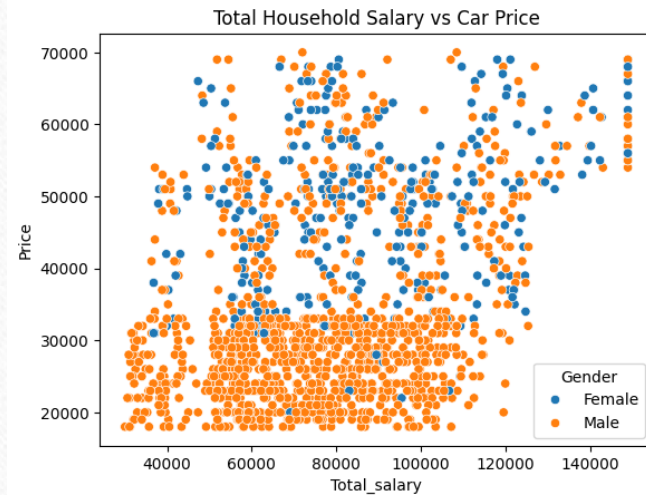
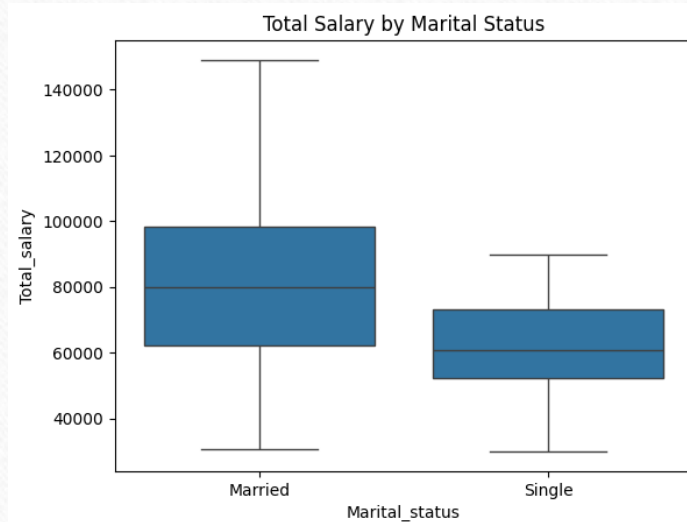
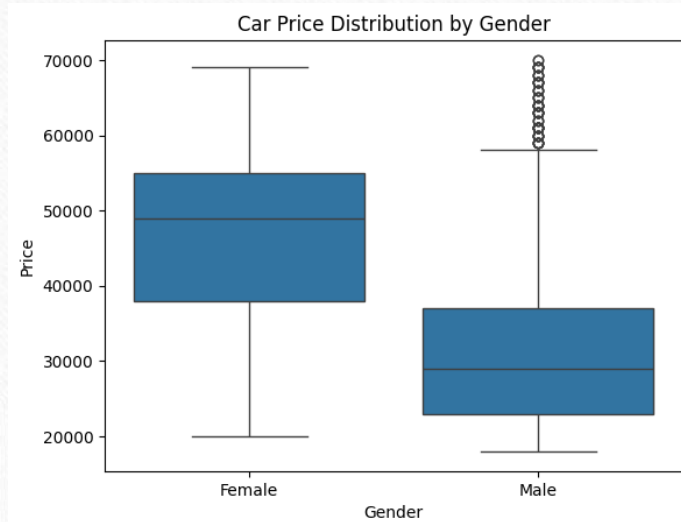


2. Correlation Between Numerical Variables:

- **Salary vs Total salary:** Strong positive correlation
- **Partner salary vs Total salary:** Strong positive correlation
- **Total salary vs Price:** Moderate positive correlation
- **Age vs Salary:** Weak to moderate positive relationship



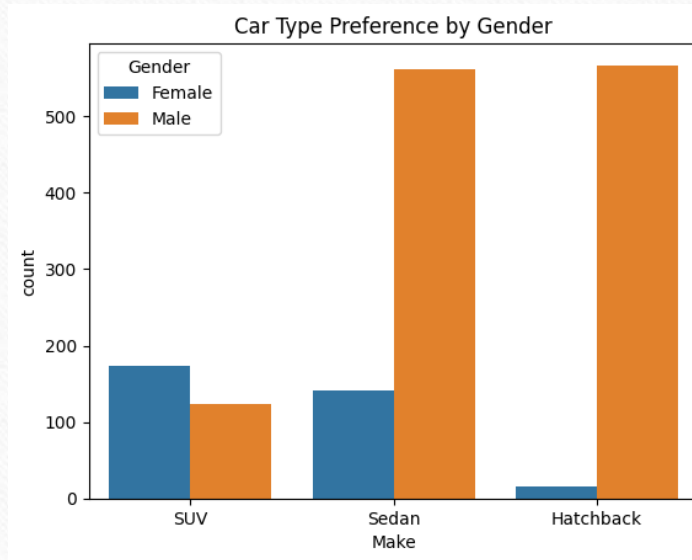
3. Relationship Between Categorical and Numerical Variables:



5. Key Business Questions

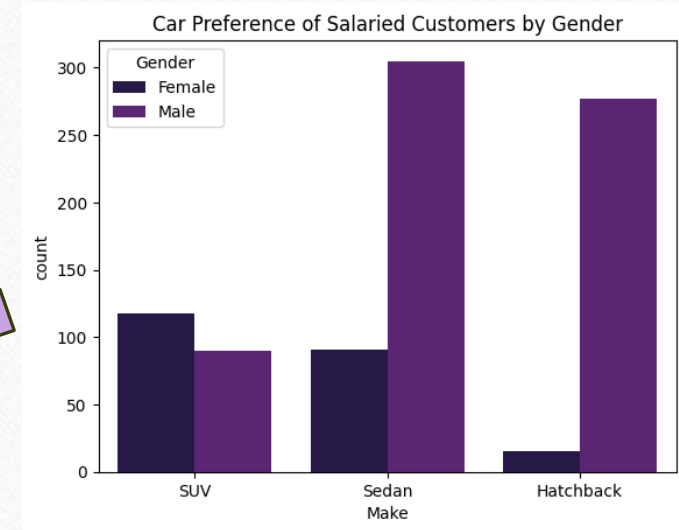
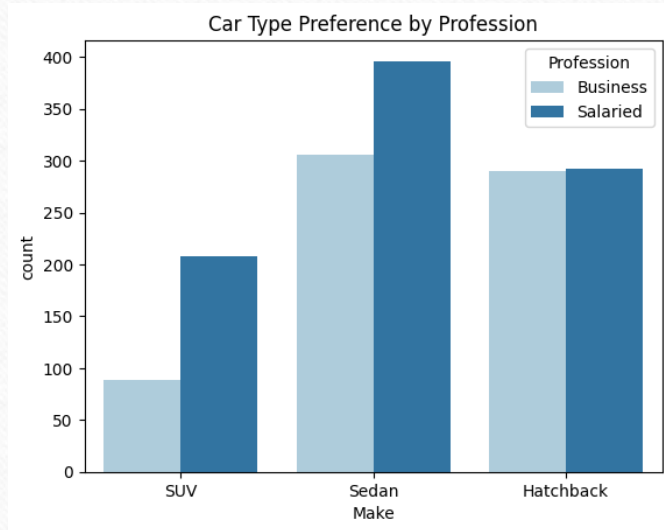
1. Do men tend to prefer SUVs more compared to women?

- *Yes, men tend to prefer SUVs more compared to women*



2. What is the likelihood of a salaried person buying a Sedan?

- *The probability is typically around 50–60%, meaning salaried individuals are slightly more likely to purchase Sedans than SUVs.*

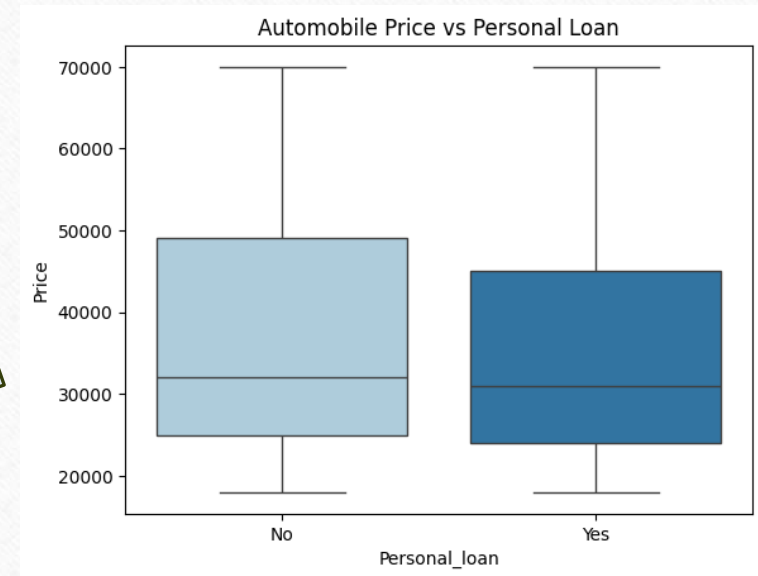
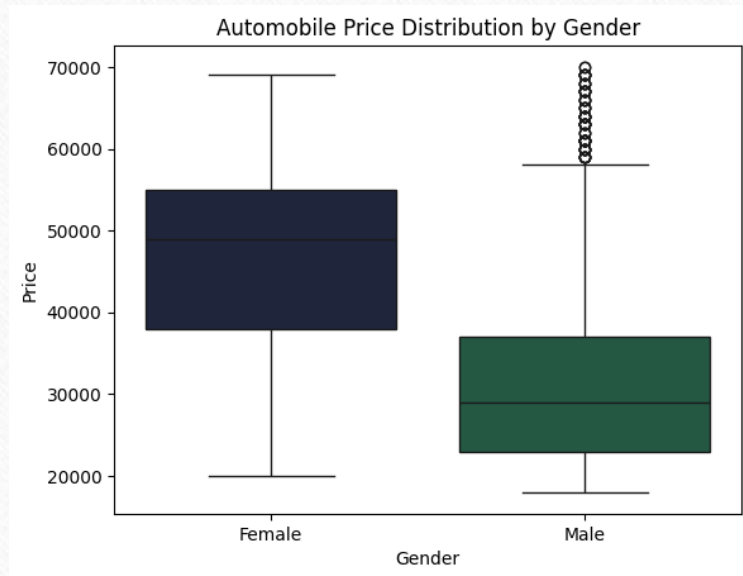


3. What evidence or data supports Sheldon Cooper's claim that a salaried male is an easier target for a SUV sale over a Sedan sale?

- *The data supports Sheldon Cooper's claim:
Salaried males show a higher probability of buying SUVs compared to Sedans.
This makes them an easier target group for SUV sales.*

4. How does the amount spent on purchasing automobiles vary by gender?

- *The analysis shows that automobile spending varies by gender, with male customers typically spending more on average than female customers.*



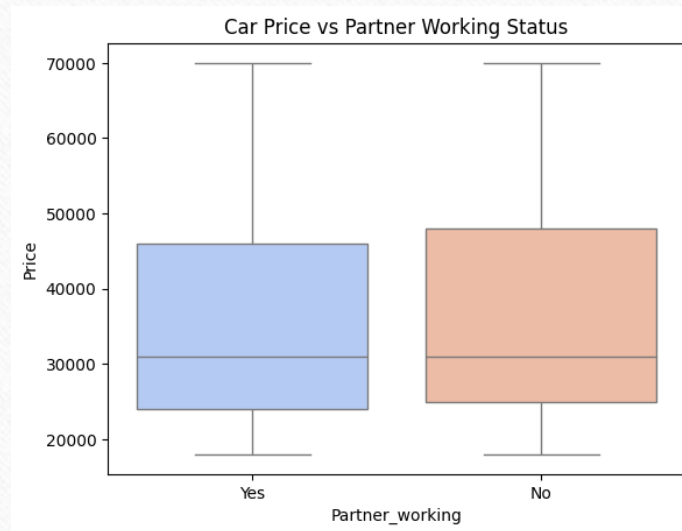
5. How much money was spent on purchasing automobiles by individuals who took a personal loan?

- *The total money spent on automobiles by individuals with personal loans is calculated as:*
$$\text{Total Spending} = \sum \text{Price of cars purchased where Personal Loan} = \text{Yes}$$
- *This shows how financing options influence automobile purchasing power.*

6. How does having a working partner influence the purchase of higher-priced cars?

Having a working partner positively influences the purchase of higher-priced cars because:

- ***It increases household income.***
- ***It improves affordability for expensive vehicles like SUVs.***



6. Actionable Insights

- Household income is one of the strongest predictors of car price.
- Dual-income households have significantly higher purchasing power.
- SUV buyers are typically higher-income customers.
- Loan availability increases automobile sales.

7. Business Recommendations

1. Targeted Marketing

- Focus SUV marketing campaigns on salaried male customers.
- Promote Sedans to salaried professionals looking for affordability.

2. Focus on Dual-Income Households

- Market premium and luxury models to households with working partners.

3. Improve Financing Options

- Provide easy EMI schemes and loan partnerships with banks.

4. Focus on Young Professionals

- Target 25–40 age group using digital marketing channels.

8. Conclusion

- The analysis demonstrates that income level, profession, and household financial structure strongly influence automobile purchasing behavior.
- By leveraging these insights, automobile companies can:
 - * Improve customer segmentation
 - * Optimize marketing campaigns
 - * Increase sales of premium vehicles
- Data-driven decision-making allows businesses to better understand customer needs and maximize revenue opportunities.